# Fruit, Vegetable and Antioxidant Intakes are Lower in Older Adults with Depression

Studies have shown an association between depression and both antioxidant levels and oxidant stress, but generally have not included intakes of antioxidants and antioxidant-rich fruits and vegetables. The present study examined the cross-sectional associations between clinically-diagnosed depression and intakes of antioxidants, fruits and vegetables in a cohort of older adults. Antioxidant, fruit and vegetable intakes were assessed in 278 elderly participants (144 with depression, 134 without depression) using a Block 1998 food frequency questionnaire, which was administered between 1999 and 2007. All participants were age 60 years or over. Vitamin C, lutein and cryptoxanthin intakes were significantly lower among depressed individuals than in comparison participants ($p<0.05$). In addition, fruit and vegetable consumption, a primary determinant of antioxidant intake, was lower in depressed individuals. In multivariable models, controlling for age, sex, education, vascular comorbidity score, body mass index, total dietary fat, and alcohol, vitamin C, cryptoxanthin, fruits and vegetables remained significant. Antioxidants from dietary supplements were not associated with depression. Antioxidant, fruit and vegetable intakes were lower in individuals with late-life depression than in comparison participants. These associations may partially explain the elevated risk of cardiovascular disease among older depressed individuals. In addition, these findings point to the importance of antioxidant food sources rather than dietary supplements.